Target dossier: BRAF, Serine/threonine-protein kinase B-raf

UniProtKB P15056 · organism: Homo sapiens · generated 2026-11-05T01:18:03Z

Data report built only from source records saved in the dossier. Interpretation is not part of this document.

Contents of the dossier

Section	Parameters	Retrieved (UTC)	File
Target profile (UniProt) and target-disease evidence (Open Targets)	max_diseases=10	2026-11-05T01:17:28Z	profile.json
Structures (RCSB PDB, AlphaFold DB)	max_models=7, mutation='V600E'	2026-11-05T01:17:31Z	structures_v600e.json
Ligands and bioactivity (ChEMBL)	mutation='V600E', max_ligands=15	2026-11-05T01:17:54Z	ligands_v600e.json
Literature (Europe PMC)	topic='V600E inhibitor', max_results=10, sort='relevance', include_preprints=False	2026-11-05T01:17:56Z	literature_v600e-inhibitor.json

Structure preparation: prep/manifest.json, created 2026-11-05T01:18:03Z (section 6).

1. Target profile (UniProt)

Source: UniProtKB P15056 · reviewed (Swiss-Prot) · annotation score 5/5 · last annotation update 2026-11-04 · retrieved 2026-11-05T01:17:28Z

Function

Protein kinase involved in the transduction of mitogenic signals from the cell membrane to the nucleus (Probable). Phosphorylates MAP2K1, and thereby activates the MAP kinase signal transduction pathway (PubMed:21441910, PubMed:29433126). Phosphorylates PFKFB2 (PubMed:36402789). May play a role in the postsynaptic responses of hippocampal neurons (PubMed:1508179)

Evidence: experimental, curator inference · PMID 1508179, PMID 21441910, PMID 29433126, PMID 36402789

Subcellular location

- Nucleus · no PubMed evidence listed
- Cytoplasm · PMID 19710016
- Cell membrane · PMID 19710016

Disease involvement curated by UniProt (8)

Disease	MIM	Evidence
Unnamed in UniProt. Note: Defects in BRAF are found in a wide range of cancers	*no record*	PMID 18974108
Colorectal cancer (CRC)	114500	PMID 12198537, PMID 21917714, PMID 23263490 (+1 more in dossier)
Lung cancer (LNCR)	211980	PMID 12460919
Familial non-Hodgkin lymphoma (NHL)	605027	PMID 14612909
Cardiofaciocutaneous syndrome 1 (CFC1)	115150	PMID 16439621, PMID 16474404, PMID 18042262 (+1 more in dossier)
Noonan syndrome 7 (NS7)	613706	PMID 19206169
LEOPARD syndrome 3 (LPRD3)	613707	PMID 19206169
Unnamed in UniProt. Note: A chromosomal aberration involving BRAF is found in pilocytic astrocytomas. A tandem duplication of 2 Mb at 7q34 leads to the expression of a KIAA1549-BRAF fusion protein with a constitutive kinase activity and inducing cell transformation	*no record*	PMID 18974108

Natural variants (47)

Variant	UniProt description	dbSNP	Evidence
T241M	in NS7; dbSNP:rs387906660	rs387906660	PMID 19206169
T241P	in CFC1 and LPRD3; dbSNP:rs387906661	rs387906661	PMID 18042262, PMID 19206169
T241R	in NS7; dbSNP:rs387906660	rs387906660	PMID 19206169
T244P	in CFC1; dbSNP:rs397507465	rs397507465	PMID 18042262
L245F	in CFC1; dbSNP:rs397507466	rs397507466	PMID 19206169
A246P	in CFC1; dbSNP:rs180177034	rs180177034	PMID 16474404, PMID 18042262 (+1 more in dossier)
Q257R	in CFC1; dbSNP:rs180177035	rs180177035	PMID 16439621, PMID 16474404 (+2 more in dossier)
Q262K	in CFC1; dbSNP:rs397507470	rs397507470	PMID 18042262
E275K	in CFC1; dbSNP:rs2129044284	rs2129044284	PMID 19206169
P301S	in dbSNP:rs34776339	rs34776339	PMID 17344846

Variant	UniProt description	dbSNP	Evidence
R462I	in CRC; dbSNP:rs180177032	rs180177032	PMID 12198537
I463S	in CRC; dbSNP:rs180177033	rs180177033	PMID 12198537
G464E	in CRC; dbSNP:rs121913348	rs121913348	PMID 12068308, PMID 12198537
G464V	in a colorectal cancer cell line; elevated kinase activity; efficiently induces cell transformation; dbSNP:rs121913348	rs121913348	PMID 12068308
G466A	in melanoma; dbSNP:rs121913351	rs121913351	PMID 12068308
G466E	in melanoma; dbSNP:rs121913351	rs121913351	PMID 12068308
G466V	in LNCR; dbSNP:rs121913351	rs121913351	PMID 12068308, PMID 12460919
S467A	in CFC1; dbSNP:rs869025606	rs869025606	PMID 16439621
F468S	in CFC1; dbSNP:rs397507473	rs397507473	PMID 16439621, PMID 18042262
G469A	in NHL; also in a lung adenocarcinoma sample; somatic mutation; elevated kinase activity; efficiently induces cell transformation; dbSNP:rs121913355	rs121913355	PMID 12068308, PMID 14612909 (+1 more in dossier)
G469E	in CFC1 and colon cancer; dbSNP:rs121913355	rs121913355	PMID 12068308, PMID 16439621 (+3 more in dossier)
G469R	in NHL; dbSNP:rs121913357	rs121913357	PMID 14612909
G469V	in a colorectal adenocarcinoma sample; somatic mutation; dbSNP:rs121913355	rs121913355	PMID 17344846
L485F	in CFC1; dbSNP:rs180177036	rs180177036	PMID 16439621, PMID 16474404 (+1 more in dossier)
K499E	in CFC1; dbSNP:rs180177037	rs180177037	PMID 16439621, PMID 16474404 (+1 more in dossier)
K499N	in CFC1; dbSNP:rs397507476	rs397507476	PMID 18042262, PMID 19206169
E501G	in CFC1; dbSNP:rs180177039	rs180177039	PMID 16439621, PMID 16474404
E501K	in CFC1; dbSNP:rs180177038	rs180177038	PMID 16439621, PMID 16474404 (+1 more in dossier)
L525P	in CFC1; dbSNP:rs869025340	rs869025340	PMID 19206169
W531C	in NS7; dbSNP:rs606231228	rs606231228	PMID 19206169
N580D	in CFC1; dbSNP:rs2128999571	rs2128999571	PMID 18042262
N581D	in CFC1; dbSNP:rs180177040	rs180177040	PMID 16439621, PMID 16474404 (+1 more in dossier)
N581S	in a colorectal adenocarcinoma sample; somatic mutation; dbSNP:rs121913370	rs121913370	PMID 17344846
E586K	in ovarian cancer; dbSNP:rs121913340	rs121913340	PMID 12068308
D594G	in NHL; dbSNP:rs121913338	rs121913338	PMID 14612909
F595L	in CFC1; also found in colon cancer; dbSNP:rs121913341	rs121913341	PMID 12068308, PMID 16439621 (+2 more in dossier)
G596R	in a colorectal adenocarcinoma sample; somatic mutation; dbSNP:rs121913361	rs121913361	PMID 12068308, PMID 17344846
G596V	in CFC1; dbSNP:rs397507483	rs397507483	PMID 16439621
L597R	in LNCR; also found in an ovarian serous carcinoma sample; somatic mutation; dbSNP:rs121913366	rs121913366	PMID 12068308, PMID 12460919 (+1 more in dossier)
L597V	in NS7; also in a lung adenocarcinoma sample; somatic mutation; elevated kinase activity; efficiently induces cell transformation; dbSNP:rs121913369	rs121913369	PMID 12068308, PMID 17344846 (+1 more in dossier)
T599R	in CFC1; dbSNP:rs121913375	rs121913375	PMID 19206169
V600D	in a melanoma cell line; requires 2 nucleotide substitutions; dbSNP:rs121913377	rs121913377	PMID 12068308
V600E	in CRC; also found in sarcoma, metastatic melanoma, ovarian serous carcinoma, pilocytic astrocytoma; somatic mutation; most common mutation; constitutive and elevated kinase activ…	rs113488022	PMID 12068308, PMID 12198537 (+6 more in dossier)
K601E	in CRC; dbSNP:rs121913364	rs121913364	PMID 12198537
K601Q	in CFC1; dbSNP:rs121913364	rs121913364	PMID 19206169
D638E	in CFC1; dbSNP:rs180177042	rs180177042	PMID 19206169
Q709R	in CFC1; dbSNP:rs397507486	rs397507486	PMID 19206169

Mutagenesis experiments (7)

Mutation	Effect reported in UniProt	Evidence
M53D	Reduces interaction with KSR1 and MAP2K1 and thus phosphorylation of MAP2K1.	PMID 29433126
K88E	Reduces interaction with KSR1 and MAP2K1 and thus phosphorylation of MAP2K1.	PMID 29433126
K483S	Reduces kinase activity with MAP2K1.	PMID 21441910
R509H	Loss of MAP2K1-mediated-BRAF-KSR1 dimerization.	PMID 29433126
K578R	Blocks EGF-induced ubiquitination and ERK activation.	PMID 23907581

Mutation	Effect reported in UniProt	Evidence
I666R	No effect on MAP2K1-mediated-BRAF-KSR1 dimerization, however loss of BRAF-mediated phosphorylation of MAP2K1.	PMID 29433126
R671K	Increased kinase activity and stability in response to EGF treatment.	PMID 21917714

Cross-references

- Ensembl gene: ENSG00000157764
- ChEMBL target: CHEMBL5145

2. Target-disease evidence and tractability (Open Targets)

Source: Open Targets Platform, BRAF (ENSG00000157764) · data release 26.09 · retrieved 2026-11-05T01:17:28Z

Tractability flags

Modality	Flag = yes	Flag = no
small molecule	Approved Drug, Structure with Ligand, High-Quality Ligand, High-Quality Pocket, Druggable Family	Advanced Clinical, Phase 1 Clinical, Med-Quality Pocket
antibody	UniProt loc high conf, GO CC high conf, Human Protein Atlas loc	Approved Drug, Advanced Clinical, Phase 1 Clinical, UniProt loc med conf, UniProt SigP or TMHMM, GO CC med conf
PROTAC	Literature, UniProt Ubiquitination, Database Ubiquitination, Half-life Data, Small Molecule Binder	Approved Drug, Advanced Clinical, Phase 1 Clinical
other clinical	none	Approved Drug, Advanced Clinical, Phase 1 Clinical

Open Targets' own yes/no flags per modality; they are not a druggability verdict.

Top 10 associated diseases by overall score

500 of 2928 associations examined (top 500 of 2928 by overall association score). Scores (0-1) aggregate evidence; they are not proof of causality.

Disease (link to evidence)	Score	Evidence types (score)
cardiofaciocutaneous syndrome	0.877	genetic association 0.94; genetic literature 0.86; affected pathway 0.83; literature 0.53; animal model 0.45; somatic mutation 0.00
Noonan syndrome	0.838	affected pathway 0.92; genetic association 0.88; genetic literature 0.82; animal model 0.44; literature 0.12
melanoma	0.820	literature 1.00; clinical 0.98; somatic mutation 0.80; genetic association 0.70; affected pathway 0.43
cardiofaciocutaneous syndrome 1	0.764	genetic association 0.94; genetic literature 0.88; animal model 0.39; literature 0.01
Noonan syndrome 7	0.759	genetic association 0.93; genetic literature 0.88; animal model 0.33; literature 0.03
LEOPARD syndrome 3	0.756	genetic association 0.89; genetic literature 0.88; animal model 0.36; literature 0.01
Noonan syndrome with multiple lentigines	0.751	genetic literature 0.82; genetic association 0.56; animal model 0.42; literature 0.11
colorectal cancer	0.749	literature 0.97; clinical 0.94; genetic association 0.77; animal model 0.31; genetic literature 0.30
cancer	0.710	literature 0.99; affected pathway 0.94; genetic association 0.43; clinical 0.42
lung cancer	0.705	genetic association 0.83; literature 0.81; genetic literature 0.61; clinical 0.18

Top 10 oncology associations

430 of the 500 examined associations belong to Open Targets' therapeutic area 'cancer or benign tumor' (its own classification, which includes some syndromes).

Disease (link to evidence)	Score	Evidence types (score)
melanoma	0.820	literature 1.00; clinical 0.98; somatic mutation 0.80; genetic association 0.70; affected pathway 0.43
colorectal cancer	0.749	literature 0.97; clinical 0.94; genetic association 0.77; animal model 0.31; genetic literature 0.30
cancer	0.710	literature 0.99; affected pathway 0.94; genetic association 0.43; clinical 0.42
lung cancer	0.705	genetic association 0.83; literature 0.81; genetic literature 0.61; clinical 0.18
non-small cell lung carcinoma	0.700	literature 0.99; somatic mutation 0.80; affected pathway 0.76; clinical 0.61
lung adenocarcinoma	0.689	affected pathway 0.95; somatic mutation 0.80; literature 0.72; clinical 0.14
hepatocellular carcinoma	0.668	clinical 0.97; literature 0.67; somatic mutation 0.46; genetic association 0.00
lymphoma	0.652	genetic association 0.86; genetic literature 0.61; somatic mutation 0.39; clinical 0.16; literature 0.14
neoplasm	0.644	literature 1.00; clinical 0.90; somatic mutation 0.46; genetic association 0.28
renal cell carcinoma	0.627	clinical 0.95; somatic mutation 0.30; literature 0.23

3. Structures

Experimental structures (RCSB PDB), variant filter 'V600E'

7 of 38 entries shown. RCSB text search for 'V600E' found 38 entries; examined the 38 best-resolution ones, 26 of which name the variant in the mutation field or title (12 discarded). Picked up to 7 from the confirmed ones: first the best-resolution entry for each distinct (mutations, nucleotide, ligand yes/no) combination, then the next best resolution. Retrieved 2026-11-05T01:17:31Z.

PDB	Title	Method	Mutations	Ligands	Nucl.	Citation
8C7Y	Crystal structure of BRAF V600E in complex with a hybrid compound 6	X-RAY DIFFRACTION 1.65 Å 2023-02-22	title: V600E	TXV (~{N}-[3-[(5-chloranyl-1~{H}-pyrrolo[2,3-b]pyridin-3-yl)ca...)	none	PubMed DOI
5JSM	BRAFV600E Kinase Domain In Complex with Chemically Linked Vemurafenib Inhibitor VEM-3-VEM	X-RAY DIFFRACTION 2.19 Å 2016-09-14	field: A688R, F667E, I543A, I544S, I551K, K630S, K723G, L588N, L706S, L716E, P722S, Q562R, Q709R, S713E, S720E, V600E, Y673S	6NB (N,N'-{ethane-1,2-diylbis[oxyethane-2,1-diyloxy-4,1-phenyl...) BEN (BENZAMIDINE) EOH (ETHANOL)	none	PubMed DOI
4CQE	B-Raf Kinase V600E mutant in complex with a diarylthiazole B-Raf Inhibitor	X-RAY DIFFRACTION 2.3 Å 2014-12-10	title: V600E other text: YES	CQE (N-{4-[2-(1-cyclopropylpiperidin-4-yl)-4-(3-{[(2,5-difluor...)	none	PubMed DOI
3OG7	B-Raf Kinase V600E oncogenic mutant in complex with PLX4032	X-RAY DIFFRACTION 2.45 Å 2010-09-22	field: A688R, F667E, I544A, I551K, K630S, L588N, L706S, L716E, Q562R, Q709R, S713E, S720E, Y673S title: V600E	032 (N-(3-{[5-(4-chlorophenyl)-1H-pyrrolo[2,3-b]pyridin-3-yl]c...)	none	PubMed DOI
4XV1	B-Raf Kinase V600E oncogenic mutant in complex with PLX7904	X-RAY DIFFRACTION 2.47 Å 2015-10-28	field: A688R, F667E, I543A, I544S, I551K, K630S, L588N, Q562R, S446A, S447A, V600E, Y673S	904 (N'-(3-{[5-(2-cyclopropylpyrimidin-5-yl)-1H-pyrrolo[2,3-b]...)	none	PubMed DOI
6V34	Crystal structure of BRAF V600E oncogenic mutant in complex with TAK-580	X-RAY DIFFRACTION 3.15 Å 2020-12-02	field: A688R, F667E, I543A, I544S, I551K, K630S, K723A, L588N, L706S, L716E, P722S, Q562R, Q709R, S713E, S720E, V600E, Y673S	QOP (6-amino-5-chloro-N-[(1R)-1-(5-{[5-chloro-4-(trifluorometh...)	none	*no record*
7ZR0	CryoEM structure of HSP90-CDC37-BRAF(V600E) complex.	ELECTRON MICROSCOPY 3.4 Å 2022-12-14	title: V600E	partner chain: Hsp90 co-chaperone Cdc37 partner chain: Heat shock protein HSP 90-beta	ATP	PubMed DOI

Not confirmed for the variant (sample): 4XV9 (B-Raf Kinase domain in complex with PLX5568), 4E26 (BRAF in complex with an organic inhibitor 7898734), 3PRF (Crystal Structure of Human B-Raf Kinase Domain in Complex w…), 3PPK (Human B-Raf Kinase in Complex with a Non-Oxime Furopyridine…), 4H58 (BRAF in complex with compound 3)

Predicted structure (AlphaFold DB)

AF-P15056-F1: mean pLDDT 66.38 (low), residues 1-766, model v6. Fraction of residues: very high (>90) 0.294, confident (70-90) 0.258, low (50-70) 0.061, very low (<50) 0.386. Canonical sequence model; it also has the highest mean pLDDT. Files: PDB CIF PAE_IMAGE

A single static prediction without ligands, nucleotides or ions.

4. Ligands and bioactivity (ChEMBL)

Ligands, variant 'V600E'

Target: CHEMBL5145 Serine/threonine-protein kinase B-raf (SINGLE PROTEIN, Homo sapiens). Retrieved 2026-11-05T01:17:54Z.

ChEMBL targets that include this protein but were not used: CHEMBL3559685 RAF serine/threonine protein kinase (PROTEIN FAMILY); CHEMBL3883317 B-raf/RAF proto-oncogene serine/threonine-protein kinase (PROTEIN FAMILY); CHEMBL4106189 BRAF/CRAF (PROTEIN COMPLEX); CHEMBL4523687 Protein cereblon/Serine/threonine-protein kinase B-raf (PROTEIN-PROTEIN INTERACTION); CHEMBL5291967 von Hippel-Lindau disease tumor suppressor/Serine/threonine-protein kinase B-raf (PROTEIN-PROTEIN INTERACTION)

Activities: 9683 in ChEMBL, 6059 fetched (truncated: most potent first, lowest pChEMBL fetched 7.23, plus every activity of mechanism compounds), 2017 used. Excluded: other or unstated mutation: 3827; potential duplicate: 194; mutation label conflicts description: 21. Minimum assay confidence: 8.

At least 1740 ligands have qualifying data for 'V600E' (download truncated: only the most potent activities were seen); returned the top 15 by best pChEMBL (listed clinical-stage first) plus 9 with a recorded mechanism. 4 more compounds with a recorded mechanism have no qualifying IC50/Ki/Kd (see mechanism_compounds_without_qualifying_activity). 1 have data only for other variants (see mechanism_compounds_other_variants_only).

Compound	Max phase	Mechanism	pChEMBL (best)	Potency (best per group)	Sources
CHEMBL3301612 ENCORAFENIB	4 (approved 2018)	Serine/threonine-protein kinase B-raf inhibitor	9.52	V600E · IC50 · binding: 0.3 nM (median 0.3, n=1) V600E · IC50 · cellular: 2 nM (median 2, n=1) [A-375]	CHEMBL3880435 (2013)
CHEMBL2028663 DABRAFENIB	4 (approved 2013)	Serine/threonine-protein kinase B-raf inhibitor	9.3	V600E · IC50 · binding: 0.5 nM (median 0.7, n=7) V600E · IC50 · cellular: 1 nM (median 6, n=6) [A-375, A375P-F11]	PMID 32781217 (2020) PMID 36758304 (2023) PMID 38954961 (2024)
CHEMBL1336 SORAFENIB	4 (approved 2005)	Serine/threonine-protein kinase B-raf inhibitor	9	V600E · IC50 · binding: 1 nM (median 25.2, n=10) V600E · IC50 · cellular: 43 nM (median 3321, n=2) [A-375, Sf9] V600E · Kd · binding: 260 nM (median 272, n=2)	PMID 18942827 (2008) PMID 29461827 (2018) PMID 23726028 (2013)
CHEMBL1229517 VEMURAFENIB	4 (approved 2011)	Serine/threonine-protein kinase B-raf inhibitor	8.49	V600E · IC50 · binding: 3.2 nM (median 26, n=13) V600E · Kd · cellular: 10 nM (median 10, n=1) [WM278] V600E · IC50 · cellular: 17 nM (median 105.5, n=4) [A-375, Malme-3M] V600E · Kd · binding: 58.2 nM (median 58.2, n=1)	PMID 23623674 (2013) PMID 27085672 (2016) PMID 25965804 (2015)
CHEMBL1946170 REGORAFENIB	4 (approved 2012)	Serine/threonine-protein kinase B-raf inhibitor	7.72	V600E · IC50 · binding: 19 nM (median 19, n=1)	PMID 22222036 (2012)
CHEMBL558752 RAF-265	2	Serine/threonine-protein kinase B-raf inhibitor	9.3	V600E · IC50 · binding: 0.5 nM (median 10.25, n=2) V600E · IC50 · cellular: 40 nM (median 40, n=4) [A-375, Malme-3M] V600E · Kd · binding: 330 nM (median 330, n=1)	PMID 26396681 (2015) PMID 30613326 (2018) PMID 18183025 (2008)
CHEMBL3545027 XL-281	2	Serine/threonine-protein kinase B-raf inhibitor	8.22	V600E · IC50 · binding: 6 nM (median 6, n=1)	PMID 29461827 (2018)
CHEMBL2029988 CEP-32496	2	Serine/threonine-protein kinase B-raf inhibitor	7.85	V600E · Kd · cellular: 14 nM (median 14, n=1) [HEK293] V600E · IC50 · cellular: 82 nM (median 82, n=1) [A-375]	PMID 22168626 (2012)
CHEMBL3577124 LY-3009120	1	Serine/threonine-protein kinase B-raf inhibitor	8.26	V600E · IC50 · binding: 5.5 nM (median 5.65, n=2) V600E · IC50 · cellular: 37 nM (median 37, n=1) [A-375]	PMID 30529543 (2019) PMID 25965804 (2015)
CHEMBL5289891 no name in ChEMBL	none recorded	*no record*	10.47	V600E · IC50 · cellular: 0.034 nM (median 0.034, n=1) [HEK293]	PMID 38954961 (2024)
CHEMBL5591799 no name in ChEMBL	none recorded	*no record*	10.18	V600E · IC50 · binding: 0.066 nM (median 0.066, n=1)	PMID 38954961 (2024)
CHEMBL5087577 no name in ChEMBL	none recorded	*no record*	10.15	V600E · IC50 · binding: 0.07 nM (median 0.07, n=1)	PMID 33999621 (2021)
CHEMBL3752213 no name in ChEMBL	none recorded	*no record*	10	V600E · IC50 · binding: 0.1 nM (median 0.1, n=1)	PMID 26739779 (2016)
CHEMBL2047875 no name in ChEMBL	none recorded	*no record*	9.96	V600E · IC50 · binding: 0.11 nM (median 0.11, n=1) V600E · IC50 · cellular: 2.5 nM (median 2.5, n=1) [MALM-3M]	PMID 22335519 (2012)
CHEMBL525191 no name in ChEMBL	none recorded	*no record*	9.89	V600E · Ki · binding: 0.13 nM (median 0.16, n=2) V600E · IC50 · binding: 0.17 nM (median 0.17, n=1) V600E · Kd · binding: 0.18 nM (median 0.18, n=1)	PMID 25267006 (2014) PMID 22808911 (2012) PMID 29461827 (2018)
CHEMBL5566280 no name in ChEMBL	none recorded	*no record*	9.89	V600E · Ki · binding: 0.13 nM (median 0.13, n=1)	PMID 39026636 (2024)
CHEMBL477989 SB590885	none recorded	*no record*	9.8	V600E · IC50 · binding: 0.16 nM (median 0.16, n=1) V600E · IC50 · cellular: 28 nM (median 58, n=3) [COLO 205, HT-29]	PMID 29461827 (2018) 10.6019/CHEMBL5444388 (2024)
CHEMBL2047879 no name in ChEMBL	none recorded	*no record*	9.74	V600E · IC50 · binding: 0.18 nM (median 1.09, n=2) V600E · IC50 · cellular: 4.6 nM (median 4.6, n=1) [MALM-3M]	PMID 22335519 (2012)
CHEMBL1668922 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 2.6, n=2)	PMID 21211972 (2011)
CHEMBL3236463 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 0.2, n=1) V600E · IC50 · cellular: 16 nM (median 16, n=1) [Malme-3M]	PMID 24675381 (2014)
CHEMBL3236470 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 0.2, n=1) V600E · IC50 · cellular: 15 nM (median 15, n=1) [Malme-3M]	PMID 24675381 (2014)
CHEMBL3236471 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 0.2, n=1) V600E · IC50 · cellular: 3 nM (median 3, n=1) [Malme-3M]	PMID 24675381 (2014)

Compound	Max phase	Mechanism	pChEMBL (best)	Potency (best per group)	Sources
CHEMBL3797403 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 0.2, n=1) V600E · IC50 · cellular: 6 nM (median 6, n=1) [A-375]	PMID 27085672 (2016)
CHEMBL3932572 no name in ChEMBL	none recorded	*no record*	9.7	V600E · IC50 · binding: 0.2 nM (median 0.2, n=1) V600E · IC50 · cellular: 1 nM (median 1, n=1) [A-375]	CHEMBL3880435 (2013)

Drugs and clinical candidates with a recorded mechanism but no qualifying IC50/Ki/Kd (4; typically antibodies or other proteins, so they are not ranked by potency):

Compound	Type	Max phase	Mechanism
CHEMBL4303729 PLIXORAFENIB	Small molecule	2	Serine/threonine-protein kinase B-raf inhibitor
CHEMBL3545080 RG-7256	Unknown	1	Serine/threonine-protein kinase B-raf inhibitor
CHEMBL3545263 LGX-806	Unknown	1	Serine/threonine-protein kinase B-raf inhibitor
CHEMBL5316167 ARQ-736 FREE ACID	Small molecule	1	Serine/threonine-protein kinase B-raf inhibitor

Compounds with a recorded mechanism and qualifying IC50/Ki/Kd only for variants other than 'V600E' (1; no qualifying measurement on this variant in ChEMBL, which is not the same as untested):

Compound	Type	Max phase	Mechanism
CHEMBL4209157 LIFIRAFENIB	Small molecule	2	Serine/threonine-protein kinase B-raf inhibitor

5. Literature (Europe PMC)

Literature, topic 'V600E inhibitor', sorted by relevance

10 of 1649 matching records. Query: (TITLE_ABS:"BRAF" OR TITLE_ABS:"Serine/threonine-protein kinase B-raf") AND (TITLE_ABS:V600E AND TITLE_ABS:inhibitor) NOT SRC:PPR. Retrieved 2026-11-05T01:17:56Z.

#	Title / authors	Journal (year)	Type	Cited by
1	Post-transplant BRAF/MEK inhibitor maintenance sustains complete remission in BRAF V600E-mutant histiocytic sarcoma. Karim I, Arnold T, Bräuninger A, Dohse M, Schell F, Samra MA, Tarawneh T, Burchert A, Gat…	Annals of hematology (2026)	Journal Article, Case Reports, case-report	0
2	Clinical and Molecular Outcomes of Dabrafenib/Trametinib-Based Combination Therapy in Acquired BRAF V600E-Mutant NSCLC Following Tyrosine Kinase Inhibitor Resistance: A Preliminary Single-Center Retrospective Study. Xu D, Hao F, Liu X, Wang X, Zhong D.	Combinatorial chemistry & high throughput screening (2026)	Journal Article	0
3	New Treatment Strategy and Future Research Direction for BRAF-Mutated Cancer. Takahashi M, Taniguchi SH, Yoshida Y.	Cancer science (2026)	review-article, Review, Journal Article	0
4	Population-Based Multicenter Real-World Outcomes of Encorafenib Plus Panitumumab in BRAF V600E-Mutated Metastatic Colorectal Cancer. Gill S, Liu N, Ngan E, Wani RJ, Cheung WY.	Clinical colorectal cancer (2026)	Journal Article	0
5	ERK1/2 Inhibitor Ulixertinib in Pancancer Patients with BRAF Fusions or Non-V600E/K Mutations: Results from the NCI-MATCH ECOG-ACRIN Trial (EAY131) Subprotocol Z1L. Subbiah V, Zhao F, Kudchadkar RR, Sullivan RJ, Chen HX, Gray RJ, Wang V, McShane LM, Rubi…	Clinical cancer research : an official journal of the American Association for Cancer Research (2026)	research-article, Journal Article	0
6	Renal and systemic involvement in Erdheim-Chester disease and targeted therapy with BRAF and MEK inhibitors Queffeulou G, Beuscart T, Leterrier N, Lechaplais Z, Potier J, Bouet J, Rauline A, Tedlao…	Nephrologie & therapeutique (2026)	Journal Article, English Abstract	0
7	Successful rechallenge of BRAF/MEK inhibitor with ruxolitinib in BRAF/MEK inhibitor-induced hemophagocytic lymphohistiocytosis like hyperinflammatory syndrome in BRAF V600E-mutated lung adenocarcinoma: a case report. Chandra S, Hansraj S, Adhikari A, Ponvilawan B, Tolentino A, Bansal D.	Frontiers in oncology (2026)	Journal Article, Case Reports, case-report	0
8	Cytotoxic activity of selinexor in BRAFV600E-driven pediatric low-grade glioma 3D cultures. Zeiser C, Emperador M, Herter S, Sigaud L, Stroh-Dege A, Loboda A, Kolodziejczak A, Zeune…	Acta neuropathologica communications (2026)	Research Support, Non-U.S. Gov't, research-article, Journal Article	0
9	Effective Treatment of Lung Adenocarcinoma With a Novel SLC44A1-BRAF Fusion Using Pembrolizumab Followed by Trametinib: A Case Report. Yasui S, Honda T, Onishi I, Ikeda S, Miyazaki Y.	Cureus (2024)	Journal Article, Case Reports, case-report	2
10	Lenvatinib plus pembrolizumab for the treatment of patients with non-BRAF mutated anaplastic thyroid cancer: A single center, phase 2 clinical trial. Cabanillas ME, Busaidy NL, Zafereo M, Iyer PC, Wang JR, Hamidi S, Ferrarotto R, Liu S, Gu…	European journal of cancer (Oxford, England : 1990) (2026)	Clinical Trial, Phase II, Journal Article	1

6. Structure preparation (Phase 2)

Manifest prep/manifest.json created 2026-11-05T01:18:03Z by prepare_structures from structures_v600e.json (retrieved 2026-11-05T01:17:31Z). Variant filter: V600E. 7 of 7 experimental structures in structures_v600e.json + the AlphaFold model. The dossier file itself holds 7 of 38 RCSB entries (see its `selection`).

Structures in the manifest

PDB	Resolution	R-free	Receptor file	Target chains (UniProt numbering)	Mutations
8C7Y	1.65 Å	0.175	clean/8C7Y_receptor.pdb	A: modelled 448-721; no internal gap; variant position: variant (GLU 600) B: modelled 449-721; missing 600-614; variant position: not modelled	title: V600E
5JSM	2.19 Å	0.2597	clean/5JSM_receptor.pdb	A: modelled 448-721; missing 602-613; variant position: variant (GLU 600) B: modelled 448-721; missing 604-615, 628-630; variant position: variant (GLU 600) C: modelled 448-722; missing 601-615, 662; variant position: variant (GLU 600) D: modelled 448-721; missing 488-489, 603-613, 628-630, 660-662; variant position: variant (GLU 600)	field: A688R, F667E, I543A, I544S, I551K, K630S, K723G, L588N, L706S, L716E, P722S, Q562R, Q709R, S713E, S720E, V600E, Y673S
4CQE	2.3 Å	0.2552	clean/4CQE_receptor.pdb	A: modelled 449-720; missing 467-468, 597-614, 629-630; variant position: not modelled B: modelled 448-720; missing 466-468, 597-614, 627-630; variant position: not modelled	title: V600E other text: YES
3OG7	2.45 Å	0.2575	clean/3OG7_receptor.pdb	A: modelled 449-720; missing 545-547, 597-614, 627-630; variant position: not modelled B: modelled 449-720; missing 601-614, 627-630; variant position: variant (GLU 600)	field: A688R, F667E, I544A, I551K, K630S, L588N, L706S, L716E, Q562R, Q709R, S713E, S720E, Y673S title: V600E
4XV1	2.47 Å	0.273	clean/4XV1_receptor.pdb	A: modelled 449-705; missing 466-468, 488, 545-546, 597-613, 627-630, 682-683; variant position: not modelled B: modelled 449-705; missing 465-469, 601-615, 627-630; variant position: variant (GLU 600)	field: A688R, F667E, I543A, I544S, I551K, K630S, L588N, Q562R, S446A, S447A, V600E, Y673S
6V34	3.15 Å	0.2859	clean/6V34_receptor.pdb	A: modelled 449-721; missing 486-487, 600-613, 629-631; variant position: not modelled B: modelled 449-721; missing 597-616, 628-630, 671-673; variant position: not modelled	field: A688R, F667E, I543A, I544S, I551K, K630S, K723A, L588N, L706S, L716E, P722S, Q562R, Q709R, S713E, S720E, V600E, Y673S
7ZR0	3.4 Å	not reported	clean/7ZR0_receptor.pdb	K (label D): modelled 521-724; missing 597-614; variant position: not modelled	title: V600E

Ligand pockets

10 pockets computed of 15 ligand instances in the 7 of 7 entries that were prepared (nucleotides, cofactors and additives are not counted as ligands; unprepared entries have no pockets). Pocket = target residues within 5 Å of the ligand instance; box = ligand + 5 Å per side, mmCIF frame.

PDB	Ligand instance	Pocket residues	Box (size, centre)	Flags	AlphaFold mean pLDDT
8C7Y	TXV U (auth A) ~{N}-[3-[(5-chloranyl-1~{H}-pyrrolo[2,3-b]pyridin-3-yl)carb…	26 residues chain A: 26	21.8 × 16.27 × 22 Å, centre (0.02, 110.789, 15.175)	none	86.32 (at 26 UniProt positions)
8C7Y	TXV X (auth B) ~{N}-[3-[(5-chloranyl-1~{H}-pyrrolo[2,3-b]pyridin-3-yl)carb…	28 residues chain B: 28	21.79 × 14.75 × 22.36 Å, centre (-6.363, 86.769, 18.087)	missing 600-614 next to the pocket (chain B)	83.73 (at 28 UniProt positions)
5JSM	6NB J (auth B) N,N'-{ethane-1,2-diylbis[oxyethane-2,1-diyloxy-4,1-phenylen…	47 residues chain A: 24 chain B: 23	19.7 × 48.88 × 18.87 Å, centre (-12.82, -1.723, 19.457)	spans several target chains large box (side > 30 Å)	87.28 (at 24 UniProt positions)
5JSM	6NB R (auth C) N,N'-{ethane-1,2-diylbis[oxyethane-2,1-diyloxy-4,1-phenylen…	47 residues chain C: 24 chain D: 23	19.21 × 48.95 × 19.89 Å, centre (-14.92, -3.056, 49.139)	spans several target chains large box (side > 30 Å)	87.28 (at 24 UniProt positions)
4CQE	CQE C (auth A) N-{4-[2-(1-cyclopropylpiperidin-4-yl)-4-(3-{[(2,5-difluorop…	29 residues chain A: 29	25.1 × 16.59 × 20.55 Å, centre (-1.516, 49.227, 21.915)	missing 597-614 next to the pocket (chain A)	85.61 (at 29 UniProt positions)
4CQE	CQE D (auth B) N-{4-[2-(1-cyclopropylpiperidin-4-yl)-4-(3-{[(2,5-difluorop…	27 residues chain B: 27	21.78 × 20.82 × 23.35 Å, centre (2.46, 50.698, -7.656)	none	87.28 (at 27 UniProt positions)
3OG7	032 C (auth A) Vemurafenib, PLX4032	22 residues chain A: 22	26.3 × 15.13 × 16.4 Å, centre (2.643, -2.28, -19.403)	missing 597-614 next to the pocket (chain A)	86.44 (at 22 UniProt positions)
4XV1	904 C (auth A) N'-(3-{[5-(2-cyclopropylpyrimidin-5-yl)-1H-pyrrolo[2,3-b]py…	22 residues chain A: 22	28.01 × 15.57 × 16.27 Å, centre (3.018, -2.32, -19.681)	missing 597-613 next to the pocket (chain A)	86.57 (at 22 UniProt positions)
6V34	QOP C (auth A) tovorafenib	25 residues chain A: 25	22.31 × 22.61 × 14.12 Å, centre (5.457, -10.185, 16.024)	none	87.52 (at 25 UniProt positions)

PDB	Ligand instance	Pocket residues	Box (size, centre)	Flags	AlphaFold mean pLDDT
6V34	QOP D (auth B) tovorafenib	25 residues chain B: 25	22.29 × 21.89 × 16.9 Å, centre (0.751, -13.762, 40.181)	none	87.52 (at 25 UniProt positions)

Ligand instances without a pocket:

- 5JSM BEN H: fewer than 10 heavy atoms
- 5JSM EOH I: fewer than 10 heavy atoms
- 5JSM EOH M: fewer than 10 heavy atoms
- 5JSM EOH N: fewer than 10 heavy atoms
- 5JSM BEN P: fewer than 10 heavy atoms

AF-P15056-F1: file raw/AF-P15056-F1-model_v6.cif; residue at UniProt 600: VAL (pLDDT 49.06)

Protein-protein interfaces

Not analysed (analyze_interfaces not run on this manifest).

Structure chosen by the user

No structure chosen (select_structure not called).

7. Methods and limitations

This document is generated deterministically from the dossier files listed at the top. It contains no interpretation: values are quoted from the source records, every link is built from an identifier returned by the source, and fields without data are shown as 'no record'. Selections are partial by design; each section states how many records were shown out of how many were found.

Selection rules and caveats stated by the tools that produced the data:

- PDB structures are a diverse selection, not the complete list.
- AlphaFold is a single static prediction without ligands, nucleotides or ions.
- All links are built from IDs/URLs returned by RCSB and AlphaFold DB.
- Ligands are listed clinical-stage first (max_phase), then by best pChEMBL: potency is not selectivity or developability, and many top values are single measurements (see n_measurements).
- ChEMBL max_phase: 4 approved, 3/2/1 clinical phase, 0.5 early phase 1, -1 clinical but phase unknown to ChEMBL, None preclinical.
- Compounds with a recorded mechanism on this target but no qualifying IC50/Ki/Kd (typically antibodies and other proteins) are listed in mechanism_compounds_without_qualifying_activity with name, type and phase: they are drugs too.
- Potency is grouped by (mutation, standard_type, assay_class); cellular and biochemical values and different mutations are never pooled. 'not stated' does NOT prove wild type.
- Activities whose curated mutation label disagrees with the mutation named in the assay text are excluded and listed in label_conflicts_flagged.
- Only compounds with a recorded mechanism or inside the top-N are checked for max_phase: other clinical-stage compounds may exist among the ligands not returned.
- Only exact '=' relations enter the statistics; censored values (>, <) are counted in activities.excluded.
- name is ChEMBL's preferred name and is None for most research compounds; no names are inferred.
- Linking PDB ligands to ChEMBL (by InChIKey) is not done here: no match means 'no record', never a guess.
- 1 compounds with a recorded mechanism have qualifying data only for other variants than 'V600E' (mechanism_compounds_other_variants_only): they are not untested; they have no qualifying measurement on this variant in ChEMBL (one may exist but be excluded: censored relation, low confidence, label conflict).
- Truncated: the 6000 most potent of 9683 activities were fetched (lowest pChEMBL fetched 7.23; activities tied at that value may be partly missing), plus every activity of the 14 compounds with a recorded mechanism. ligands_with_qualifying_data is therefore a lower bound. For other compounds, n_measurements, median and worst values only count measurements above that cutoff, and weaker ligands are not seen. With a mutation filter, activities of that variant below the cutoff are also missed.
- Search matches the gene symbol or protein name in title/abstract; ambiguous symbols can bring unrelated papers, so check titles.
- Only abstracts are returned, not full text.
- All links are built from IDs returned by Europe PMC.

Structure preparation rules (prep/manifest.json, decisions):

- Original files are kept unmodified in raw/. Loop completion, protonation and charges are Phase 3.
- Receptor (clean/<PDB>_receptor.pdb): target chains only (RCSB mapping), first model, first alternate conformation, no hydrogens, no waters, no additives, no ligands; nucleotides, cofactors and metal ions MG/MN/ZN/FE/CO/NI/CU that touch a target chain are KEPT (e.g. KRAS GDP + Mg, 8TVK); NA, K, CL, CA, CD are removed. Coordinates are unchanged. PDB format unless chain names or atom count do not fit, then mmCIF.
- Pocket: target-chain residues with any atom within 5 Å of the ligand instance (gemmi, first alternate conformation of the ligand), in the asymmetric unit only: crystal-symmetry copies are excluded (they put residues ~47 Å away in 8C7Y). UniProt numbers come from label_seq_id + RCSB alignment; residues of other polymer chains are listed apart. No pocket for nucleotides and cofactors (kept in the receptor), for ligands with fewer than 10 heavy atoms, or without a target-chain neighbour in RCSB: each is listed in pocket_not_computed. Box: ligand bounding box + 5 Å per side, same frame as the mmCIF; sides above 30 Å are flagged. A ligand instance with pocket residues in more than one target chain is flagged spans_target_chains (e.g. a bivalent compound bridging two sites): its pocket and box cover every chain it touches and are NOT split into one box per site; residues_by_chain gives the count per chain.
- Chain metrics: missing_uniprot_ranges are residues of the sample sequence not modelled inside the modelled span; a missing range next to a pocket residue is flagged. variant_check reads the residue actually modelled at the variant position in each target chain and compares it with the UniProt canonical letter (not the letter typed in the filter): the variant can be named in the title and still be unmodelled in the chain that holds the ligand (3OG7 A). Two residues at one position (microheterogeneity) give status 'ambiguous'.
- AlphaFold pLDDT (B-factor column, equal to _ma_qa_metric_local) is read at the same UniProt numbers as each experimental pocket, only for the canonical-sequence model; residues without a UniProt number are not compared (alphafold_residues_compared vs n_residues).
- Target chains come from RCSB's chain-to-UniProt mapping, not from the mmCIF _struct_ref, which is unreliable (3OG7 cites UniProt Q5IBP5; antibody chains often cite no UniProt).
- Chains are given as label_asym_id (mmCIF) and auth_asym_id (author, used by PyMOL and the PDB format); they can differ (7ZR0: BRAF is label D, auth K).
- Ligand poses are the crystal coordinates of each instance (RCSB ModelServer, by label_asym_id). A ligand can sit on one chain only (3OG7: chain A) or touch several.
- neighbor_*_chains come from RCSB rcsb_target_neighbors with the closest distance per chain; RCSB does not document the distance cutoff.
- construct.mutations are the depositor's declared mutations of the target entity; they can include solubility/engineering mutations besides the variant (3OG7 declares 13). Any declared mutation means the structure is not the native sequence; an empty list does NOT prove it is native (the variant may be named only in the title, or the text may be unparsed, e.g. 'YES').
- Additives (ions, buffers, cryoprotectants) get no SDF. The classification is the one of get_structures; some additives (e.g. BEN, EOH in 5JSM) are not in its list and count as ligands.
- Glycans (RCSB branched entities, e.g. NAG chains) are not queried and are absent from nonpolymer_instances.

Not done yet (pending): step 4: user's structure choice (select_structure).